Cognitive Modeling

Complete Lecture Notes

Week 10: Bayesian Modeling II: Posterior Predictive Checks and Bayes Factors

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Chapter 1

Bayesian Modeling

1.1 Week 10: Bayesian Modeling II: Posterior Predictive Checks and Bayes Factors

> Chapter 4: Bayesian modeling (with Chapter 3: evaluation) · Lecture 1 h · Lab 2 h Running data: the Week 2 forgetting data (exponential vs. power), and a weak-evidence SDT participant (is $d' = 0$?).

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
posterior	توزیع پسین (posterior)	باور درباره‌ی پارامتر بعد از دیدن داده
forgetting curve	منحنی فراموشی	کاهش یادآوری با گذشت زمان
exponential / power function	توانی / تابع نمایی	دو شکل رایج برای منحنی فراموشی و یادگیری
evidence	شواهد	اطلاعاتی که برای تصمیم جمع می‌شود
sensitivity (d′)	حساسیت	فاصله‌ی توزیع علامت و نویز
posterior predictive check (PPC)	بررسی پیش‌بینی پسین	مقایسه‌ی داده‌ی شبیه‌سازی‌شده از پسین با داده‌ی واقعی
marginal likelihood / evidence	درست‌نمایی حاشیه‌ای	احتمال داده تحت مدل، میانگین روی پیشین
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها

English	فارسی	توضیح کوتاه
Bayes factor (BF)	عامل بیز (Bayes factor)	نسبت درست‌نمایی‌های حاشیه‌ای دو مدل
model flexibility / complexity	پیچیدگی مدل / انعطاف‌پذیری	دامنه‌ی الگوهایی که مدل می‌تواند تولید کند
grid approximation	تقریب شبکه‌ای (grid)	محاسبه‌ی پسین روی شبکه‌ای از مقادیر
Savage–Dickey ratio	نسبت Savage–Dickey	BF از نسبت چگالی پسین به پیشین در مقدار صفر
AIC / BIC	بیزی / معیار اطلاعاتی آکائیکه	برازش به‌علاوه‌ی جریمه برای تعداد پارامتر
prior	توزیع پیشین (prior)	باور درباره‌ی پارامتر پیش از دیدن داده
estimate / estimator	برآوردگر / برآورد	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
posterior odds	بخت پسین (odds)	نسبت احتمال پسین دو مدل
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
noise	نویز	تغییرات تصادفی در شواهد یا در داده
adequacy check	بررسی کفایت مدل	آیا مدل ویژگی‌های اصلی داده را بازتولید می‌کند
nested models	مدل‌های تودرتو (nested)	یک مدل حالت خاص مدل دیگر است
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
accuracy	دقت	نسبت پاسخ‌های درست

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Generate posterior predictive distributions and conduct graphical and numerical posterior predictive checks (PPCs).
2. Define the marginal likelihood and the Bayes factor, and explain why the marginal likelihood penalizes model flexibility automatically.
3. Compute Bayes factors by grid integration and with the Savage–Dickey density ratio, and compare them with the BIC approximation.
4. Evaluate the sensitivity of Bayes factors to the prior, and contrast it with the robustness of posterior estimates.
5. Interpret Bayes factors on a graded evidence scale and combine them with prior odds into posterior model probabilities.

1.1.3 Motivation & big picture

Last week gave us posteriors *within* a model. Two questions remain.

1. Is the model any good at all? A posterior can be narrow and confident even when the model is badly wrong: the exponential model will happily give a precise posterior for λ for the Week 2 data. The Bayesian answer is to ask the fitted model to generate new data and check whether they look like the real data. This is a *posterior predictive check*, the Bayesian counterpart of residual plots and G^2.

2. Which model do the data support more? In Week 7 we compared models with AIC and BIC, which penalize parameter counts. Bayesian inference has a built-in answer: compute how well each model predicted the data *before* it was fitted, averaging over all parameter values its prior allows. That is the marginal likelihood, and the ratio of two marginal likelihoods is the Bayes factor. A flexible model spreads its predictions over many possible datasets, so it assigns less probability to any one of them. The Bayes factor thus implements Occam's razor automatically, and it responds to functional-form flexibility, not just to k.

Bayes factors are also used to test null hypotheses ("is sensitivity zero?"). Unlike p-values, they can quantify evidence for the null.

Where is this used?

- Evidence for "no effect". Replication studies often ask whether an effect is absent. A p-value cannot support the null; a Bayes factor can ("the data are 8 times more likely under no effect").
- Psychology journals and software: Bayes factors are reported routinely with tools such as JASP and the R package BayesFactor, for t-tests, ANOVAs and correlations as well as for cognitive models.
- Model checking in every Bayesian analysis: posterior predictive checks are a standard step before any result is trusted (e.g. in clinical trials analyzed with Bayesian methods and in cognitive modeling papers).
- Comparing theories with different flexibility: when two models have the same number of parameters but one can produce many more data patterns, the Bayes factor is one of the few tools that notices.

1.1.4 Lecture notes

In plain words Two separate questions this week. (1) Is my model good enough? Let the fitted model *imagine* new data: take plausible parameter values from the posterior, simulate a new experiment, repeat many times. If the real data look like a typical imagined dataset, the model is adequate; if the real data sit outside the cloud of imagined data, the model is missing something. This is a

posterior predictive check. (2) Which of two models do the data support more? Ask each model, *before fitting*, how probable it found the data we actually got, averaging over all the parameter values its prior allows. That number is the marginal likelihood, and the ratio for two models is the Bayes factor. A model that bets on everything (very flexible) must spread its bets thinly, so it wins less when the data fall where a simpler model bet. That is Occam's razor, built into probability theory. The catch: the Bayes factor depends on the prior, so you must report how it changes with reasonable priors.

Suggested timing (60 min): posterior predictive checks 15 · marginal likelihood and Bayes factors 20 · computing BFs (grid, BIC, Savage–Dickey) 15 · prior sensitivity and comparison of methods 10.

1. Posterior predictive checks

The posterior predictive distribution describes new data y^{rep} that the fitted model expects:

$$p(\mathrm{y}^{\text{rep}} \mid \mathrm{y}) = \int p(\mathrm{y}^{\text{rep}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathrm{y})\, d\boldsymbol{\theta}.$$

Simulation recipe: for $s = 1, \ldots, S$, draw $\boldsymbol{\theta}^{(s)}$ from the posterior and simulate a dataset $\mathrm{y}^{\text{rep}(s)}$ from the model with the real design. This propagates *both* parameter uncertainty and sampling noise.

Checks: - Graphical: plot the observed data on top of the predictive intervals (e.g., the central 95% of y^{rep} per condition). Systematic departures (observed points repeatedly outside the bands, or on one side of them) show where the model fails. - Numerical: choose a test statistic $T(\mathrm{y})$ that captures a feature you care about (e.g., recall at the longest delays, the steepness of the early drop). The posterior predictive p-value is $p_B = P\big(T(\mathrm{y}^{\text{rep}}) \geq T(\mathrm{y}) \mid \mathrm{y}\big)$. Values near 0 or 1 indicate that the model cannot reproduce this feature.

PPCs assess absolute adequacy, i.e., whether the model is good enough, not which of two models is better. They are a routine step of every Bayesian analysis (Week 13).

Reading the math

- y^{rep}: "replicated" data, an imagined repetition of the experiment.
- $\int p(\mathrm{y}^{\text{rep}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathrm{y})\, d\boldsymbol{\theta}$: average the model's predictions over all parameter values, each weighted by its posterior plausibility. In practice this integral is replaced by the two-step simulation recipe: draw $\boldsymbol{\theta}$ from the posterior, then simulate data with it.
- $p_B = P(T(\mathrm{y}^{\text{rep}}) \geq T(\mathrm{y}) \mid \mathrm{y})$: the fraction of imagined datasets whose summary T is at least as large as the real one. If only 1% of imagined datasets show recall as high as observed at 30 days ($p_B = 0.01$), the model cannot produce that feature.

2. The marginal likelihood and the Bayes factor

The marginal likelihood (evidence) of model $\mathcal{M}$ is the likelihood averaged over the prior:

$$p(\mathrm{y} \mid \mathcal{M}) = \int p(\mathrm{y} \mid \boldsymbol{\theta}, \mathcal{M})\, p(\boldsymbol{\theta} \mid \mathcal{M})\, d\boldsymbol{\theta}.$$

It is the probability the model assigned to the observed data *before seeing them*. Bayes' rule at the level of models gives:

$$\underbrace{\frac{p(\mathcal{M}_1 \mid \mathrm{y})}{p(\mathcal{M}_2 \mid \mathrm{y})}}_{\text{posterior odds}} = \underbrace{\frac{p(\mathrm{y} \mid \mathcal{M}_1)}{p(\mathrm{y} \mid \mathcal{M}_2)}}_{\mathrm{BF}_{12}} \times \underbrace{\frac{p(\mathcal{M}_1)}{p(\mathcal{M}_2)}}_{\text{prior odds}}.$$

BF_{12} is the factor by which the data change the odds in favor of $\mathcal{M}_1$ over $\mathcal{M}_2$, and $\mathrm{BF}_{21} = 1/\mathrm{BF}_{12}$.

The automatic Occam's razor. Both models must spread a total probability of 1 over all possible datasets. A simple model concentrates its predictions on a few datasets; a flexible model spreads them thin. If the data fall where the simple model predicted, the simple model gets the higher marginal likelihood, even though the flexible model can also fit them well after its parameters are tuned. Unlike AIC/BIC, this penalty depends on *how* flexible a model is (the prior predictive spread), not merely on k.

A common evidence scale (an adaptation of Jeffreys' classic categories, as used by Lee & Wagenmakers):

BF_{12}	Evidence for $\mathcal{M}_1$
1–3	Anecdotal
3–10	Moderate
10–30	Strong
30–100	Very strong
> 100	Extreme

(Values below 1 are read the same way for $\mathcal{M}_2$.) These labels are aids, not thresholds. Report the number.

Reading the math

- $p(\mathrm{y} \mid \mathcal{M}) = \int p(\mathrm{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$: the likelihood averaged over the prior, not maximized. Compare: MLE takes the *best* parameter value; the marginal likelihood takes the *prior-weighted average* over all of them. A model with many bad parameter values in its prior gets a low average.
- Posterior odds = Bayes factor × prior odds. Odds are probability ratios: odds of 3 mean 3:1, i.e. probability 0.75. If both models were equally plausible before (prior odds = 1) and $\mathrm{BF}_{12} = 3$, then afterwards $P(\mathcal{M}_1 \mid \mathrm{y}) = 3/(1+3) = 0.75$.
- Worked example in one line: with 7 of 10 correct, chance predicted the data with probability 0.117 and the "any accuracy" model with 0.091, so $\mathrm{BF}_{01} = 1.29$: the data hardly change the odds.

3. Computing Bayes factors

(a) Grid integration (1–2 parameters): $p(\mathrm{y} \mid \mathcal{M}) \approx \sum_g p(\mathrm{y} \mid \boldsymbol{\theta}_g)\, p(\boldsymbol{\theta}_g)\, \Delta_g$, where Δ_g is the cell area. Do it in log space (log-sum-exp), and check that the prior mass inside the grid is ≈ 1.

(b) The BIC approximation. For large N, $\ln p(\mathrm{y} \mid \mathcal{M}) \approx -\frac{1}{2}\mathrm{BIC}$, so

$$\mathrm{BF}_{12} \approx \exp\big(-\tfrac{1}{2}(\mathrm{BIC}_1 - \mathrm{BIC}_2)\big).$$

It is quick but crude: it ignores the actual prior (it implicitly assumes a "unit-information" prior) and functional-form flexibility.

(c) The Savage–Dickey density ratio (nested models). If $\mathcal{M}_0$ fixes $\phi = \phi_0$ in $\mathcal{M}_1$, and the priors on the remaining parameters match, then

$$\mathrm{BF}_{01} = \frac{p(\phi = \phi_0 \mid \mathrm{y}, \mathcal{M}_1)}{p(\phi = \phi_0 \mid \mathcal{M}_1)}\,:$$

the posterior density at the null value divided by the prior density at the null value, both under the larger model. If the data make ϕ_0 more plausible than it was a priori, the evidence favors the null. One posterior from $\mathcal{M}_1$ is all we need.

(Further methods, such as bridge sampling and thermodynamic integration, compute marginal likelihoods from MCMC output for complex models. They are beyond this course.)

Reading the math Savage–Dickey $\mathrm{BF}_{01} = p(\phi_0 \mid \mathrm{y})/p(\phi_0)$: look at the height of the posterior and of the prior at the null value ϕ_0 (e.g. $\theta = 0.5$). If the posterior is *higher* there than the prior was, the data made the null value more plausible, so the evidence favors the null. In the worked example the prior (flat) has height 1 at 0.5 and the posterior $\mathrm{Beta}(8, 4)$ has height 1.29, so $\mathrm{BF}_{01} = 1.29$. A picture of two curves and one vertical line at ϕ_0 is all you need to explain this.

4. Prior sensitivity: estimation vs. testing

For estimation, a reasonable prior matters less and less as data accumulate. For Bayes factors, the prior on a tested parameter *never* stops mattering. Making the prior under $\mathcal{M}_1$ wider spreads its predictions more thinly, which lowers $p(\mathrm{y} \mid \mathcal{M}_1)$, so an arbitrarily vague prior makes the BF favor the null no matter what the data are (the Jeffreys–Lindley paradox). Consequences: - Priors for model comparison must be chosen thoughtfully, ideally reflecting realistic effect sizes. - Always report a sensitivity analysis: the BF over a range of reasonable prior widths.

5. Comparison of model-evaluation methods

Method	Question	Penalizes	Prior needed?	Week
G^2, residuals, PPC	Is the model adequate?	—	PPC: yes	4, 10

Method	Question	Penalizes	Prior needed?	Week
LRT	Does the extra parameter improve fit beyond chance? (nested only)	Δk via χ^2	No	7
AIC / CV	Which model predicts new data best?	Estimated over-fitting	No	6–7
BIC	Approximately: which model has the higher evidence?	$k \ln N$	Implicit	7
Bayes factor	How much do the data change the odds between models?	Prior predictive spread (true flexibility)	Yes, and the result depends on it	10

> Advanced: Bayes factors vs. predictive criteria. The marginal likelihood can be written as a product of sequential predictions, $p(\mathrm{y}) = p(y_1)\, p(y_2 \mid y_1) \cdots p(y_N \mid y_{1:N-1})$. It therefore scores how well the model predicts each data point from the earlier ones, *starting from the prior*. LOO-CV scores predictions from (almost) all the other data. This is why BFs depend on the prior and LOO hardly does, and why they can disagree on the same data.

For an engineer

- Posterior odds = BF × prior odds is the MAP (Bayes) detector between two hypotheses, where the Bayes factor plays the role of the likelihood ratio.
- The difference from the GLRT (Week 4 and 7): with unknown parameters, the GLRT *maximizes* the likelihood over them; the Bayes factor *integrates* (averages) over them with a prior. Maximizing favors flexible hypotheses; averaging penalizes them automatically.
- The Jeffreys–Lindley paradox has an engineering flavor: if you let the alternative hypothesis allow absurdly large signal amplitudes (a very wide prior), almost any observation is better explained by "noise only".
- Posterior predictive checks are like validating a channel model by simulating received signals from the fitted model and comparing their statistics (histograms, autocorrelations, eye diagrams) with measured ones.

For the instructor

- Explain Occam's razor with a betting story. Two forecasters each have 100 chips to place on possible outcomes. One spreads them over everything; the other puts most chips on a narrow range. If the outcome lands in that narrow range, the focused forecaster wins more. The marginal likelihood is the payout.
- Show prior sensitivity live. In the lab code, change the width of the prior under $\mathcal{M}_1$ and watch the BF move. This is more convincing than any warning.
- Keep the two questions apart. Write "Is the model good enough? $\rightarrow$ PPC" and "Which model is better? $\rightarrow$ BF / CV / AIC" on the board. Students often use a BF to claim a model is *good*; it only says it is *better than the other*.
- Typical questions.
  - *"BF or p-value?"* They answer different questions. The BF compares two specific hypotheses and can support the null; the p-value only measures surprise under the null.
  - *"BF = 2.5, is that significant?"* There is no threshold; it is anecdotal evidence. Report the number.
  - *"Why do BF and AIC disagree?"* They aim at different targets (evidence vs. prediction) and the BF depends on the prior.

1.1.5 Worked example

Part 1: is recall at chance? A Bayes factor by hand

A participant chooses between two alternatives on $n = 10$ trials and is correct on $y = 7$. Is performance at chance?

- $\mathcal{M}_0$: $\theta = 0.5$ (no free parameters).
- $\mathcal{M}_1$: $\theta \sim \text{Beta}(1, 1)$ (any accuracy equally plausible a priori).

Step 1. $p(y \mid \mathcal{M}_0) = \binom{10}{7} 0.5^{10} = 120/1024 = 0.1172$.

Step 2. Under a uniform prior, every count $0, \ldots, n$ is equally likely a priori, so $p(y \mid \mathcal{M}_1) = \int_0^1 \binom{10}{7} \theta^7 (1-\theta)^3 \, d\theta = \frac{1}{n+1} = \frac{1}{11} = 0.0909$.

Step 3. $\text{BF}_{01} = 0.1172/0.0909 = 1.29$. The data slightly *favor chance*, which is anecdotal evidence either way.

Step 4 (Savage–Dickey check). The posterior under $\mathcal{M}_1$ is $\text{Beta}(8, 4)$, whose density at 0.5 is 1.289. The prior density at 0.5 is 1. The ratio is 1.29, identical to Step 3.

More data:

Data	p (two-sided binomial test)	BF_{01}	Reading
7/10	0.34	1.29	Anecdotal, either way

Data	p (two-sided binomial test)	BF_{01}	Reading
60/100	≈ 0.06	1.10	Still anecdotal: the data barely distinguish chance from "some accuracy"
70/100	< 0.001	0.0023 ($\mathrm{BF}_{10} \approx 430$)	Extreme evidence against chance

Interpretation. With 60/100, a p-value near .05 tempts researchers to claim an effect, yet the Bayes factor says the data are about equally likely under both hypotheses. The reason: the alternative with a uniform prior "predicted" that accuracy could be anywhere from 0 to 1. An observed accuracy of 0.6 is only modestly better predicted by it than by exactly 0.5. A narrower, more realistic prior under $\mathcal{M}_1$ (e.g., accuracies between 0.5 and 0.8) would give a different BF, which is why prior sensitivity must be reported.

Part 2: posterior model probabilities

For 7/10 with prior odds of 1: posterior odds $= 1.29$, so $P(\mathcal{M}_0 \mid y) = 1.29/2.29 = 0.56$. If a reviewer believes a priori that chance performance is 4 times less likely than above-chance performance (prior odds 1:4), then posterior odds $= 1.29 \times 0.25 = 0.32$, so $P(\mathcal{M}_0 \mid y) = 0.24$. The BF is the same for everyone; the posterior model probabilities depend on the prior odds.

1.1.6 Lab (2 hours)

Goals

1. Run graphical and numerical posterior predictive checks for the exponential and power forgetting models.
2. Compute marginal likelihoods and the Bayes factor for power vs. exponential by grid integration; compare with BIC.
3. Test $d' = 0$ for a weak-evidence SDT participant with the Savage–Dickey ratio, and verify it by direct computation.
4. Map the sensitivity of the Bayes factor to the prior, and contrast it with the posterior estimate.

Timing: Part A 35 min · Part B 25 min · Part C 30 min · Part D 20 min · Part E 10 min.

Setup

Code · کد: `code/week10.py`, cell 1 · سلول 1

Expected output:

```
exp: posterior mean A = 0.655, rate = 0.049
pow: posterior mean A = 0.867, rate = 0.417
```

Part A: posterior predictive checks (35 min)

Code · کد: `code/week10.py`, cell 2 · سلول 2

Expected output:

```
exp: T(observed) = 0.267, mean T(rep) = 0.240, posterior predictive p =
↪  0.302
pow: T(observed) = 0.267, mean T(rep) = 0.243, posterior predictive p =
↪  0.320
```

Interpret: look at the plots first. For the exponential model, the observed recall at the shortest delay (42/50) lies *above* the 95% band and recall at 7 days (13/50) lies *below* it. The exponential cannot drop steeply early and then level off. The power model's band contains every observed point.

Yet the numerical check with T = mean recall at the three longest delays gives unremarkable p_B values for both models (about 0.3). The exponential's misfit is in the *shape* (early vs. middle delays), which this statistic does not look at. A PPC can only detect the failures its test statistic is sensitive to. Choose statistics that target the features that distinguish the theories; in Exercise 2 a "first drop" statistic exposes the exponential model ($p_B \approx 0.02$).

Part B: Bayes factor by grid integration (25 min)

Code · کد: `code/week10.py`, cell 3 · سلول 3

Expected output:

```
exp: log p(y|M) = -34.938   (prior mass inside grid = 0.9996)
pow: log p(y|M) = -22.872   (prior mass inside grid = 0.9991)

BF(pow vs exp) = exp(12.07) = 1.74e+05
BIC approximation: BF(pow vs exp) = exp(9.98) = 2.16e+04
```

Interpret: BF $\approx 1.7 \times 10^5$ is extreme evidence for the power model over the exponential. BIC gives the same verdict but a different number (2×10^4). Why do the grid BF and the BIC approximation differ? (Hint: which prior does each use, and are the two models equally flexible under our priors?)

Part C: Savage–Dickey: is this participant's sensitivity zero? (30 min)

A participant in a hard detection task: 12 hits out of 20 signal trials, 8 false alarms out of 20 noise trials. $\mathcal{M}_1$: $d' \sim \mathcal{N}(0,1)$, $c \sim \mathcal{N}(0,1)$. $\mathcal{M}_0$: $d' = 0$, $c \sim \mathcal{N}(0,1)$.

> Code · کد: `code/week10.py`, cell 4 · سلول 4

Expected output:

```
BF01 Savage-Dickey = 1.328;  BF01 direct = 1.328;  posterior mean d' =
 ↪  0.444
MLE d' = 0.507
```

The two computations agree, which is a useful check that both are implemented correctly.

Part D: prior sensitivity (20 min)

> Code · کد: `code/week10.py`, cell 5 · سلول 5

Expected output:

```
 prior SD of d'   BF01  BF10  posterior mean d'
           0.25  0.938 1.066              0.144
           0.50  0.972 1.028              0.313
           1.00  1.328 0.753              0.444
           2.00  2.314 0.432              0.496
           5.00  5.541 0.180              0.513
          10.00 11.011 0.091              0.515
```

What to notice: as the prior on d' widens, the posterior mean of d' quickly stabilizes, but BF_{01} keeps growing in favor of the null. A very wide prior "predicts" huge sensitivities that did not occur, so the alternative is penalized. For estimation the prior washes out; for testing it does not.

Part E: interpret (10 min)

Write 4–5 sentences: does this participant detect the signal? Report the BF with your chosen prior and justify the prior, mention the sensitivity analysis, and state what additional data would make the answer clearer.

Exercises (increasing difficulty)

Exercise 1 ⋆. Verify by numerical integration (`scipy.integrate.quad`) that $\int_0^1 \binom{10}{7}\theta^7(1-\theta)^3 d\theta = 1/11$.

Exercise 2 ⋆. In Part A, use a different test statistic: the drop in recall from the first to the second interval. Which model, if any, fails this check? (Expected: the exponential, $p_B \approx 0.02$; the power model passes.)

Exercise 3 ⋆⋆ (prior sensitivity of the forgetting BF). Repeat Part B with the rate priors widened (scale $\times$ 3) and narrowed (scale $\div$ 3). Does the conclusion "power $\gg$ exponential" survive? Why is it so robust here, unlike in Part D?

Exercise 4 ⋆⋆ (an informed alternative). In Part C, replace the prior on d' by a half-normal $d' \sim \mathcal{N}^+(0, 1)$ (only positive values; remember to renormalize). Compute the Savage–Dickey BF. Why does a directional prior change the result?

Exercise 5 ⋆⋆⋆ (Advanced: BF for the repetition study). For the Week 8 case study ($\mathcal{M}_0$ vs. $\mathcal{M}_1$), choose priors for δ_0, δ_1 and κ, compute BF_{10} with the Savage–Dickey ratio at $\delta_1 = 0$ using a 3-D grid (keep it coarse), and compare it with the BIC approximation.

Lab check-off

Show: (1) the two PPC plots with p_B values, (2) the grid BF for power vs. exponential and its BIC approximation, and (3) the prior-sensitivity table from Part D with one sentence of interpretation.

1.1.7 Connection to AI & LLMs

Do LLMs have a "theory of mind"? Some studies report that LLMs pass false-belief tasks, classic tests of whether one understands that others can hold false beliefs. Others show that small changes to the stories make performance collapse [verify: Kosinski, 2023/2024; Ullman, 2023]. Many such tests have two answer options, so chance is 50%, and the number of items is small. A Bayes factor answers the right first question: *do the data support above-chance performance at all, compared with chance?*

> Code · كد: `code/week10.py`, cell 6 · سلول 6

Expected output:

```
26/40 correct (65%): BF10 = 1.16
130/200 correct (65%): BF10 = 785.26
```

The same 65% is no evidence at all with 40 items, and overwhelming evidence with 200 items. Above-chance performance is also only the first step: the posterior predictive check of this week asks *how* the model succeeds and fails. For example, does it fail on exactly the altered stories, where a human would not?

Try it (no API needed). - ⋆ Repeat with a prior concentrated on above-chance values, e.g. Beta(5, 2) restricted to $\theta > 0.5$ (renormalize the density). How does BF_{10} for 26/40 change? - ⋆⋆ Simulate a "shortcut" model that answers correctly on 90% of standard stories and 30% of altered stories (half of each). Its overall accuracy is 60%. Design a posterior predictive check that reveals the shortcut. - ⋆⋆ Discussion: is "above chance on false-belief tasks" a claim at Marr's computational or algorithmic level?

1.1.8 Common mistakes & misconceptions

1. "A narrow posterior means the model is right." Precision is not adequacy. Always run a PPC.
2. Using the posterior predictive check to choose between models. PPCs check adequacy. Use BFs, CV or information criteria to compare.
3. "The Bayes factor is the posterior probability of the model." It is the *change* in odds. Posterior probabilities also need prior odds.
4. Using very vague priors in Bayes-factor tests. They bias the BF toward the null without limit (Part D). Use realistic priors and report the sensitivity.
5. Savage–Dickey with mismatched priors. The nuisance-parameter priors must be the same in both models, and the prior density must be evaluated *at the null value* in the larger model.
6. Computing marginal likelihoods on a grid that cuts off prior mass. Check that the prior mass inside the grid is close to 1.
7. Treating "BF = 2.9" vs. "BF = 3.1" as qualitatively different. The evidence scale is continuous; labels are aids, not decision thresholds.

1.1.9 Check your understanding

1. How is a posterior predictive distribution simulated, and what does it combine? Answer Draw parameters from the posterior, then simulate data from the model with those parameters (and the real design). It combines uncertainty about the parameters with the sampling noise of new data.

2. Define the marginal likelihood in words. Answer The probability (density) of the observed data under the model, averaged over all parameter values weighted by the prior: how well the model predicted these data before it was fitted.

3. Why does the marginal likelihood penalize flexible models even without counting parameters? Answer Probability over possible datasets must sum to 1. A flexible model spreads its prior predictions over many datasets and so assigns each one, including the observed one, less probability, unless the data fall where only the flexible model predicts them.

4. $\text{BF}_{10} = 6$ and the prior odds are 1:3 (against $\mathcal{M}_1$). What are the posterior odds and $P(\mathcal{M}_1 \mid \text{y})$? Answer Posterior odds $= 6 \times 1/3 = 2$, so $P(\mathcal{M}_1 \mid \text{y}) = 2/3 \approx 0.67$.

5. State the Savage–Dickey ratio and the conditions for using it. Answer $\text{BF}_{01} = p(\phi_0 \mid \text{y}, \mathcal{M}_1)/p(\phi_0 \mid \mathcal{M}_1)$: the posterior over the prior density at the null value under the larger model. It requires nested models (the null fixes $\phi = \phi_0$) and identical priors on the other parameters.

6. Why does widening the prior on the tested parameter increase BF_{01}, even with fixed data? Answer A wider prior assigns more probability to large effects and less to the small effect sizes compatible with the data, so the alternative's marginal likelihood decreases while the null's does not change.

7. BIC gives $\text{BF}_{12} \approx 20$, while the grid computation with your priors gives 3. Which do you report and why? Answer The grid BF, because it uses the actual priors and accounts for functional-form flexibility; BIC is a large-sample approximation with an implicit prior. Mention the discrepancy and, ideally, a prior sensitivity analysis.

1.1.10 Assignment

A4: A Bayesian analysis by grid (1.0 point of 20; released Week 10, due at the start of the Week 12 lecture)

Data: the A2 dataset (`rng=np.random.default_rng(2024)`, 9 intervals, $n = 20$ items; see Week 4).

Tasks.

1. Priors (1.5 pts). Choose and justify priors for (A, λ) of the exponential model and (A, d) of the power model. Show prior predictive simulations of forgetting curves for both models and revise the priors if needed.
2. Posterior (2 pts). Compute the grid posteriors for both models. Report posterior means, SDs and 95% HDIs, and the posterior correlation of the two parameters. Plot the joint posteriors.
3. Posterior predictive checks (2 pts). Graphical PPC for both models, plus a numerical PPC with a test statistic you choose and justify.
4. Bayes factor (2 pts). Compute $\text{BF}_{\text{pow,exp}}$ by grid integration (check the prior mass), compare it with the BIC approximation, and repeat with two alternative prior settings (sensitivity analysis).
5. Comparison with A2 (1 pt). Compare the posterior summaries with your A2 MLEs and their uncertainty.
6. Interpretation (1.5 pts, max. 300 words). What do you conclude about forgetting in these data, and how robust is the conclusion to the prior?

Rubric: points as above; within each task: correctness 60%, figures and tables 20%, interpretation 20%. Late submissions lose 10% per day (max. 3 days).

1.1.11 Further reading

- Lee & Wagenmakers (2014), chapter 7. Bayesian model comparison: marginal likelihoods, Bayes factors and the Savage–Dickey method, with worked examples. *(Main reading.)*
- Farrell & Lewandowsky (2018), chapter 11. Bayesian model comparison using Bayes factors, in the context of cognitive models.
- Kruschke (2014), chapter 10. Model comparison and hierarchical modeling, a gentle introduction to Bayes factors and their prior sensitivity.
- Martin, Kumar & Lao (2021), chapter 2. Exploratory analysis of Bayesian models, including posterior predictive checks.

Preview for Week 11: our grids worked for two parameters. For five, ten or a hundred parameters we need a smarter way to explore the posterior: Markov chain Monte Carlo. You will build your first sampler from scratch.